Table 1: **Privileged object coordinates, with and without.** *State input*: the backward tower (B for FB/RLDP, ϕ for PSM/HILP) consumes the full observation. *Object coords*: it consumes the ϵ-scaled object-coordinate block slice instead — 3 numbers (cube xyz) on `cube-single`, 6 on `cube-double` — while the forward map, the left encoder and the policy keep the whole observation. Only the backward tower's input changes, so each pair is a single-knob comparison. Superscripts give the number of seeds; mean ± sd, and n=1 cells should not be quoted.

	`cube-single`		`cube-double`	
Method	state input	object coords	state input	object coords
Density-FB	$\mathbf{0.91 \pm 0.02}^{3}$	$\mathbf{0.94 \pm 0.01}^{3}$	0.05 ± 0.01^{3}	$\mathbf{0.64 \pm 0.04}^{3}$
Density-RLDP	0.23 ± 0.06^{3}	0.82 ± 0.06^{2}	—	0.17^{1}
Density-PSM	0.17 ± 0.07^{3}	0.88^{1}	—	—
Density-HILP	0.15 ± 0.07^{3}	0.15 ± 0.01^{2}	0.04^{1}	0.04^{1}
FB*	0.11^{1}	0.34^{1}	0.02^{1}	$0.22^{1,\dagger}$
RLDP	0.11 ± 0.01^{3}	0.53 ± 0.35^{2}	0.04^{1}	0.16^{1}
PSM	0.12 ± 0.03^{3}	0.94^{1}	—	—
HILP	0.08^{1}	0.09^{1}	0.03^{1}	0.06^{1}

Three readings. (i) Density-FB is the only method that works at state input (0.91 on `cube-single`, 4× the best retrofit), so transplanting the categorical estimator into RLDP/PSM/HILP does *not* reproduce it — and the missing ingredient is not the object coordinates. (ii) Object coordinates rescue the retrofits on `cube-single` (3–8×) but move Density-FB only 0.91 → 0.94; on `cube-double` the reverse holds — Density-FB needs them (0.05 → 0.64) while the retrofits reach only ∼0.2. (iii) HILP is inert to both.

***Provisional, and still unexplained.** The FB row runs `ortho_coef`= 1.0 (the value our yaml copied from td_jepa's OGBench `BASE_CFG`), which a line-by-line audit identified as the sole divergence from the torch implementation — the objective, z-law and actor all transcribe faithfully, and td_jepa's `sweep_cube` sweeps $\{100, 1000\}$, never 1.0. Rerunning at `ortho_coef`= 1000 raises `cube-single` from 0.11 to $\mathbf{0.203 \pm 0.046}$ (n=3) — real, but far short of the 0.42–0.53 of the reference stack and the 0.520 ± 0.055 of *Is Manipulation Hard for Successor Measures?*, and on `cube-double` it gives 0.000. So `ortho_coef` was one cause and not the cause. Both FB columns above are at 1.0, so the pair is internally matched; a matched `ortho_coef`= 1000 pair has not been run. Note also that the comparison targets evaluate at 1M steps against this table's 500k. †Read at 450k: the cell died at 465k with `CUDA_ERROR_STREAM_CAPTURE_INVALIDATED` (the known stream-capture fault; CUDA graphs are disabled in later waves). "—" marks a cell that was never run.

Harness check. Density-FB's own cells reproduce its published numbers inside this harness at this protocol (0.94 here vs 0.915 ± 0.098 in Table **??**; 0.64 vs 0.674 ± 0.051), which is what licenses reading the rest of the column.